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United States Patent	12398621
Kind Code	B2
Date of Patent	August 26, 2025
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System and apparatus for unloading well stimulation pumps

Abstract

An example system includes a fluid end having a block, a suction bore formed in the block, a suction valve seat disposed within the suction bore, and a suction valve disposed within the suction bore, the suction valve, in a closed position, being configured to contact the suction valve seat to seal the suction bore. The system also includes a rod configured to engage the suction valve, the rod being moveable between a first position and a second position, wherein in the first position the suction valve is in the closed position, and in the second position, engagement of the rod with the suction valve causes the suction valve to be in an open position.

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Appl. No.: 17/729925

Filed: April 26, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230340859 A1	Oct. 26, 2023

Publication Classification

Int. Cl.: E21B34/14 (20060101); E21B43/26 (20060101); E21B47/008 (20120101); F04B1/053 (20200101); F04B15/00 (20060101)

U.S. Cl.:

CPC E21B34/14 (20130101); E21B43/26 (20130101); E21B43/2607 (20200501);
E21B47/008 (20200501); F04B1/053 (20130101); F04B15/00 (20130101);

Field of Classification Search

CPC: E21B (34/14); E21B (43/26); E21B (43/2607); E21B (47/008); F04B (1/053); F04B (15/00); F04B (49/03)

USPC: 417/53

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1365272	12/1920	Reeder	417/294	F04B 49/243
1395090	12/1920	Carpenter	417/27	H02P 1/04
1527954	12/1924	Holdsworth	137/596.2	F04B 49/00
1602938	12/1925	Riesner	417/27	F04B 49/02
1633035	12/1926	Reck	417/446	F04B 49/03
2937801	12/1959	Berkoben	417/281	F04B 49/03
3077836	12/1962	Coberly	251/324	F04B 53/144
3459363	12/1968	Miller	417/446	F04B 49/243
3773438	12/1972	Hall	417/345	E21B 43/2607
5356269	12/1993	Stangroom	417/505	F04B 49/243
8366408	12/2012	Wago	417/295	F04B 53/1032
8506262	12/2012	Leugemors	417/311	F04B 53/1022
9677558	12/2016	Van Himme	N/A	F02M 59/48
10190718	12/2018	Weinstein et al.	N/A	N/A
10550836	12/2019	Shampine et al.	N/A	N/A
10683742	12/2019	Elmer	N/A	N/A
10850970	12/2019	Schmidt	N/A	N/A
2001/0036413	12/2000	McFadden	417/293	E02F 9/2239
2003/0010025	12/2002	Evans	60/431	F16H 61/468
2007/0065302	12/2006	Schmitz	417/298	F04B 39/08
2008/0085180	12/2007	Mackenzie	415/31	F04C 28/08
2009/0178387	12/2008	Schultz	60/274	F02D 19/0642
2010/0183448	12/2009	Leugemors	417/53	F04B 49/22
2014/0331659	12/2013	Kato	60/434	B66F 9/22
2014/0361541	12/2013	Todd	415/4.2	F03D 15/00
2020/0182236	12/2019	Beisel	N/A	F04B 1/053
2020/0208628	12/2019	Sigmar	N/A	F04C 11/008
2020/0256333	12/2019	Surjaatmadja	N/A	F04B 49/03
2022/0341202	12/2021	Goettl	N/A	E04H 4/12
2023/0332592	12/2022	He	N/A	F04B 49/22

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a pump system. More specifically, the present disclosure relates to a mechanism for maintaining an open position of a valve in a fluid end to unload a well stimulation pump during start up or wind down of a prime mover that drives the well stimulation pump.

BACKGROUND

(2) Hydraulic fracturing is a well stimulation technique that typically involves pumping hydraulic fracturing fluid into a wellbore at a specific rate and pressure necessary to form fractures in a rock formation surrounding a targeted region of the wellbore. This well stimulation technique often enhances the natural fracturing of a rock formation in order to increase the permeability of the rock formation, thereby improving recovery of oil, natural gas, and/or other fluids. For example, such techniques are also performed to enhance recovery of water in water wells.

(3) In order to fracture such rock formations, the hydraulic fracturing fluid is injected into the wellbore at high pressures. Typically, a series of pumps are used to achieve such high-pressure injection of the hydraulic fracturing fluid. The series of pumps may be powered by prime movers (e.g., diesel engines, electric motors, natural gas engines, etc.). During a hydraulic fracturing process, several pumps are pumping at the same time with several prime movers powering the pumps. If one of the pumps being used encounters a problem and needs to be shut down, a backup pump can be started to maintain a flow rate of hydraulic fracturing fluid being pumped into the wellbore. Due to the high pressure of the hydraulic fracturing fluid being pumped into the wellbore, prime movers often lack the necessary torque to start a backup pump when the prime mover is operating at a slow speed. Furthermore, with electric motors, starting the backup pump against such a high pressure load leads to a high electrical need which can generate high levels of heat, which can cause additional problems to arise.

(4) An example hydraulic fracturing system is described in U.S. Pat. No. 10,190,718 (hereinafter referred to as "the '718 reference"). In particular, the '718 reference describes an accumulator assembly for a hydraulic fracturing system that is fluidly connectable to a flow line between a blender and a fracturing pump of the hydraulic fracturing system. The '718 reference describes that the accumulator assembly includes a pressurizable tank that contains a pressurized fluid and a control valve fluidly connected to a discharge end of the pressurizable tank and the flow line. The '718 reference describes that the control valve is opened to fluidly connect the pressurizable tank to the flow line when a pressure on the flow line is less than a target pressure. The control valve, described in the '718 reference, is closed when the pressure on the flow line is greater than or substantially the same as the target pressure. However, the '718 reference describes preventing hydraulic fracturing fluid from flowing from a fluid end back to a fluid reservoir. Therefore, the hydraulic fracturing system, described in the '718 reference, maintains a load on the hydraulic fracturing pumps, unless a hydraulic fracturing process is stopped. As such, in order to start prime movers during the hydraulic fracturing process, large amounts of fuel are required to start the prime mover against such a high load and electrical motors would require high power requirements. Furthermore, starting or winding down the prime movers under such a high load could damage or reduce a useful life of a prime mover and/or pumps used in the hydraulic fracturing process.

(5) Example embodiments of the present disclosure are directed toward overcoming the deficiencies described above.

SUMMARY

(6) An example system includes a fluid end having a block, a suction bore formed in the block, a suction valve seat disposed within the suction bore, and a suction valve disposed within the suction

bore, the suction valve, in a closed position, being configured to contact the suction valve seat to seal the suction bore. The system also includes a rod configured to engage the suction valve, the rod being moveable between a first position and a second position, wherein in the first position the suction valve is in the closed position, and in the second position, engagement of the rod with the suction valve causes the suction valve to be in an open position.

(7) An example method includes determining to start a prime mover of a fluid system. The method also includes causing one or more suction valves of a fluid end to be positioned in an open position, wherein the one or more suction valves permit fluid to flow from respective pump chambers of the fluid end to a fluid manifold when the one or more suction valves are in the open position. The method may further include receiving, from one or more sensors, sensor data from the fluid system, and determining, based at least in part on the sensor data, that operation of the fluid system satisfies a threshold. The method further includes causing, based at least in part on determining that operation of the fluid system satisfies the threshold, the one or more suction valves to be positioned in a closed position, wherein the one or more suction valves prevent fluid to flow from the respective pump chambers to the fluid manifold when the one or more suction valves are in the closed position.

(8) In a further example, a system includes a fluid end having a block, a suction bore formed in the block, a suction valve seat disposed within the suction bore, and a suction valve disposed within the suction bore, the suction valve configured to contact the suction valve seat in a closed position to seal the suction bore. The system further includes a controller configured to position the suction valve in the open position or allow it to be in the closed position, wherein the controller is configured to position the suction valve in the open position for an amount of time, the suction valve being substantially static in the open position for the amount of time.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a partial schematic view of an example fluid system having an unloading mechanism, in accordance with an example of the present disclosure.

(2) FIG. 2 is a partial schematic view of another example fluid system having an unloading mechanism, in accordance with an example of the present disclosure.

(3) FIG. 3 is a cross-sectional view of an example fluid end with an unloading mechanism in a first position, in accordance with an example of the present disclosure.

(4) FIG. 4 is a cross-sectional view of an example fluid end with an unloading mechanism in a second position, in accordance with an example of the present disclosure.

(5) FIG. 5 is a cross-sectional view of an example fluid end with an alternate embodiment of an unloading mechanism in a first position, in accordance with an example of the present disclosure.

(6) FIG. 6 is a cross-sectional view of an example fluid end with an alternate embodiment of an unloading mechanism in a second position, in accordance with an example of the present disclosure.

(7) FIG. 7 is a flowchart illustrating a method of unloading a well stimulation pump, in accordance with an example of the present disclosure.

DETAILED DESCRIPTION

(8) FIG. 1 depicts an example fluid system **100**. The fluid system **100** shown in FIG. 1 is part of a well service, workover, or well stimulation system for an oil and gas well. For example, the fluid system **100** is implemented in a hydraulic fracturing system, which is a well stimulation technique that involves pumping hydraulic fracturing fluid into a wellbore at a rate and pressure sufficient to form fractures in a rock formation surrounding the wellbore. In some examples, the fluid system **100** shown in FIG. 1 forms a portion of a hydraulic fracturing system or other well service system

and may include additional and/or alternative components than the components shown and described in FIG. 1.

(9) In some examples, the fluid system **100** includes at least one prime mover **102** coupled to a pump **104**. The prime mover **102** is coupled to the pump **104** and is configured to drive operation of the pump **104**. As such, the prime mover **102** may also be referred to herein as a “driver.” In some examples, the prime mover **102** may be a diesel engine, natural gas engine, or other type of internal combustion engine. Alternatively, the prime mover **102** may be an electric motor. In some examples, the prime mover **102** may include one or more sensors **106** that sense and generate driver speed data. The driver speed data may be sent to a controller **108** of the fluid system **100**. The one or more sensors **106** may also generate and communicate load data or other types of data to the controller **108**.

(10) In some examples, the fluid system **100** may include multiple controllers **108** that receive data from the fluid system **100** and are configured to control at least a portion of the operations of the fluid system **100** automatically and/or with user input, as will be described further herein. While the description herein may describe a single controller **108**, it is to be understood that multiple controllers **108** may be used to perform the actions described herein.

(11) In some examples, the controller **108** may be a manually operated and/or actuated controller. For example, the controller **108** may include a push button actuator, switch (e.g., pneumatic switch, electronic switch, mechanical switch, etc.), lever, or other manually operated controller that is configured to control at least a portion of the operations of the fluid system **100**. In some examples, and as described further herein below, a user (such as an operator) may operate the controller **108** which may, in turn, be configured to operate an unloading mechanism **142** (described further herein below) of the fluid system **100**. While the controller **108** is described as being configured to control at least a portion of the operations of fluid system **100**, it is to be understood that a user may manually control such operations of the controller **108**.

(12) Additionally, or alternatively, the controller **108** may be, for example, a hardware electronic control module (ECM) or other electronic control unit (ECU). The controller **108** includes, for example, a microcontroller, one or more processors, memory (e.g., RAM), storage (e.g., EEPROM or Flash) configured to perform the described functions of the controller **108**. The controller **108** controls at least a portion of the operations of the fluid system **100** automatically and/or with user input. Instead of, or in addition to, an ECM/ECU the controller **108** may include a general computer microprocessor configured to execute computer program instructions (e.g., an application) stored in memory to perform the disclosed functions of the controller **108**. As mentioned, the controller **108** includes a memory, a secondary storage device, processor(s), and/or any other computing components for running an application. Various other circuits may be associated with controller **108** such as power supply circuitry, signal conditioning circuitry, solenoid driver circuitry, actuator driver circuitry, or other circuitry. In some examples, the controller **108** and/or a portion of components of the controller **108** may be located remotely from the fluid system **100** and may be communicatively coupled to the fluid system **100**.

(13) As mentioned previously, the controller **108** may receive various types of data from components of the fluid system **100**. Furthermore, the controller **108** may provide instructions to the various components of the hydraulic fracturing system **100**. For example, the controller **108** may receive driver speed data from a driver speed sensor **106** of the prime mover **102**. The controller **108** may also provide operational instructions to the prime mover **102**, and in some examples, the controller **108** may generate such instructions based on information received from the sensor **106** and/or based on other information (e.g., one or more control maps, algorithms, or rules stored in the memory noted above). In any of the examples described herein, the controller **108** may be configured to control operation of the prime mover **102** during various tasks performed by the fluid system **100**. For example, the controller **108** may be configured to start up or wind down the prime mover **102**.

(14) The prime mover **102** may be directly or indirectly coupled to the pump **104** and may be configured to drive the pump **104**. In some examples, the pump **104** may be a hydraulic fracturing pump (or other type of well service or workover pump). The pump **104** may include various types of high-volume hydraulic fracturing pumps such as triplex pumps, quintuplex pumps, or other types of hydraulic fracturing pumps. Additionally, and/or alternatively, the pump **104** includes other types of reciprocating positive-displacement pumps or gear pumps. A number of pumps implemented in the fluid system **100** and designs of the pump **104** (or pumps) may vary depending on the fracture gradient of the rock formation that will be hydraulically fractured, the number of pumps **104** used in a fluid system **100**, the flow rate necessary to complete the hydraulic fracture, the pressure necessary to complete the hydraulic fracture, etc. The fluid system **100** includes any number of pumps **104** in order to pump hydraulic fracturing fluid at a predetermined rate and pressure. The exact configuration of the fluid system **100** varies from site to site.

(15) The pump **104** includes at least one plunger **110** that is at least partially disposed within a fluid end **112**. In some examples, the pump **104** includes multiple plungers **110** disposed within the fluid end **112**. When the pump **104** is operating, the pump **104** drives the plunger **110** in reciprocating motion. For example, the pump **104** moves the plunger **110**, at least partially within the fluid end **112**, in a first direction **114** and a second direction **116** that is opposite the first direction **114**. In some examples, the pump **104** may be configured to move the plunger **110** in reciprocal directions in order to allow fluid into the fluid end **112** and pump the fluid out of the fluid end **112**. For example, when the pump **104** moves the plunger **110** in the first direction **114**, the plunger **110** allows fluid to flow through a suction valve **118** (shown further in FIG. 3) into the fluid end **112** from a fluid source **120**. The fluid source **120** is fluidly connected to a fluid intake manifold **122** configured to distribute hydraulic fracturing fluid to one or more chambers within the fluid end **112**. Additionally, or alternatively, the fluid source **120** includes one or more water tanks, blenders, hydration units, liquid additive systems, etc.

(16) Furthermore, when the pump **104** moves the plunger **110** in the second direction **116**, the plunger **110** displaces the fluid, increasing a pressure of the fluid, until the fluid reaches a predetermined pressure. Once the fluid reaches the predetermined pressure, a discharge valve **124** (shown further in FIG. 3) within the fluid end **112** opens and allows the fluid to move from the fluid end **112** into a fluid discharge manifold **126**. In some examples, the fluid discharge manifold **126** receives pressurized fluid from one or more fluid ends and directs the fluid to a wellhead **128** where the fluid is injected into a wellbore.

(17) In some examples, the fluid end **112** includes a block **130** having one or more bores (or fluid passages) formed in the block **130** of the fluid end **112**. The block **130** may be formed from stainless steel, carbon steel, or other material. In some examples, the fluid end **112** includes a suction bore **132** formed in the block **130**. The suction bore **132** provides a fluid passageway for fluid to enter the fluid end **112** when the plunger **110** moves in the first direction **114**. The suction bore **132** includes the suction valve **118** disposed within the suction bore **132**. The suction valve **118** is configured to control flow of fluid into the fluid end **112**. For example, when the plunger **110** moves in the first direction **114**, the movement of the plunger **110** causes the pressure of the fluid in the pump chamber **140** to be lower than the pressure of the fluid in the suction bore **132** which causes the suction valve **118** to open, thereby allowing fluid into the fluid end pump chamber **140**. Conversely, when the plunger **110** moves in the second direction **116**, the suction valve **118** remains closed, allowing the plunger **110** to displace the fluid within the fluid end **112**.

(18) The fluid end **112** also includes a discharge bore **134** formed in the block **130**. The discharge bore **134** provides a fluid passageway for fluid to be discharged from the fluid end **112** to the fluid discharge manifold **126** (or other component). The discharge bore **134** includes a discharge valve **124** disposed within the discharge bore **134**. The discharge valve **124** is configured to control fluid flow from the fluid end **112**. For example, the discharge valve **124** remains closed until the fluid has reached a predetermined pressure as the plunger **110** displaces the fluid by moving in the

second direction **116**. Once the fluid reaches the predetermined pressure, the pressurized fluid causes the discharge valve **124** to open, allowing the fluid to be discharged from the fluid end **112**. (19) The fluid end **112** further includes a plunger bore **136** formed in the block **130**. The plunger bore **136** is sized to receive the plunger **110** of the pump **104** at least partially therein. As mentioned previously, the plunger **110** is moveable in the first direction **114** and the second direction **116** within the plunger bore **136** to allow fluid into the fluid end **112** via the suction bore **132** and discharge the fluid from the fluid end **112** via the discharge bore **134**. The fluid end **112** further includes a suction cover bore **138** formed in the block **130** of the fluid end **112**. The suction cover bore **138** remains sealed during operation of the pump **104**. However, the suction cover bore **138** provides access to the plunger **110** or portions of the fluid end **112** for maintenance or other reasons, while the pump **104** is not operating.

(20) The fluid end **112** also includes a pump chamber **140** disposed between the suction bore **132** and the discharge bore **134**. The pump chamber **140** is a chamber formed in the fluid end **112** that is formed at least in part by a convergence of portion(s) of the suction bore **132**, the discharge bore **134**, the plunger bore **136**, and the suction cover bore **138**. In some examples, the plunger **110** displaces the fluid in the pump chamber **140** until it reaches a predetermined pressure, which causes the discharge valve **124** to open, allowing the fluid to exit the pump chamber **140** via the discharge valve **124**.

(21) The fluid system **100** further includes an unloading mechanism **142**. The unloading mechanism **142** may be configured to maintain the suction valve **118** in an open position while the prime mover **102** starts up, winds down, or otherwise operates. Maintaining the suction valve **118** in the open position while the pump **104** operates permits fluid to circulate between the pump chamber **140** and the fluid intake manifold **122**. Additionally, the unloading mechanism **142** may prevent fluid flow through an outlet (e.g., outlet **318** in FIG. 3) of the fluid end **112** while the prime mover **102** is still winding down. The unloading mechanism **142** may include an actuator **144** coupled to a rod **146** and configured to move the rod **146** between various positions. For example, the actuator **144** may extend the rod **146** which may press against the suction valve **118**, thereby lifting the suction valve **118** and positioning the suction valve **118** in an open position. Conversely, the actuator **144** may retract the rod **146** such that the rod **146** allows the suction valve **118** to be positioned in a closed position. The actuator **144** may include an electronic actuator, a hydraulic actuator, a pneumatic actuator, or other type of actuator configured to actuate the rod **146**. In some examples, the unloading mechanism **142** is communicatively coupled to the controller **108**. For example, the controller **108** may be configured to control actuation of the actuator **144**. As such, the controller **108** may be configured to position the suction valve **118** in an open position or a closed position via actuation of the actuator **144**. In some examples, the controller **108** sends one or more signals to the actuator **144** which causes the actuator **144** to extend or retract the rod **146**, thereby causing the suction valve **118** to open or close.

(22) In some examples, the system **100** may include multiple unloading mechanisms **142** that correspond with a number of suction valves **118** in the fluid end **112**. In other words, the system **100** may include an unloading mechanism **142** for each suction valve **118** in the fluid end **112**. In some examples, individual unloading mechanisms **142** may be configured to position individual suction valves **118** independent of other unloading mechanisms **142**. However, operation of the unloading mechanisms **142** may be coordinated such that the unloading mechanisms **142** open and close the suction valves **118** sequentially or simultaneously. In some examples, the unloading mechanisms **142** As such, the unloading mechanisms **142** are configured to close or open the suction valves **118** in a specific sequence in order to reduce a rate of load being applied to the prime mover **102** during operation of the pump **104**. Control of the unloading mechanisms **142**, either in sequence or substantially synchronous, may be controlled by the controller **108**. As described further herein, the controller **108** may include an electronic controller and/or the controller **108** (e.g., such as a lever, switch, or other manually controlled apparatus) may be

manually controlled.

(23) In some examples, the controller **108** receives driver speed data from the one or more sensors **106** of the prime mover **102**. The driver speed data is indicative of a driver speed (e.g., engine or motor speed) of the prime mover **102**. The controller **108** may be configured to control operation of the unloading mechanism **142** based at least in part on the driver speed data. For example, the controller **108** may be configured to position the suction valve **118** in an open position when the driver speed data indicates that the driver speed is below a threshold driver speed. However, in some examples, the controller **108** is configured to position the suction valve **118** in an open position or a closed position irrespective of the driver speed. For example, a user may manually operate the controller **108**, thereby causing the controller **108** to position the suction valve in the open position or the closed position. As mentioned previously, while the controller **108** is described as positioning the suction valve **118** in the open position or the closed position, it is to be understood that a user may manually control such operations. In some instances, when the driver speed is below a threshold driver speed, the prime mover **102** may be starting up or winding down. As such, the controller **108** may open the suction valve **118** in order to unload the pump **104**. For example, opening the suction valve **118** while the prime mover **102** starts up or winds down allows the prime mover **102** to start or wind down under a reduced load. For example, rather than having to pump fluid to a predetermined pressure to discharge fluid through the discharge valve **124**, the fluid is circulated between the pump chamber **140** and the fluid intake manifold **122**. In some examples, the controller **108** may open the suction valve **118** for an amount of time associated with a time between starting the prime mover **102** and the driver speed of the prime mover **102** reaching the threshold driver speed.

(24) In some examples, the controller **108** is configured to allow the suction valve **118** to move to a closed position when the driver speed data indicates that the driver speed is substantially equal to or greater than the threshold driver speed. In some instances, once the driver speed reaches the threshold driver speed, the prime mover **102** may be capable of driving operation of the pump **104** at a required torque, pump speed, pump rate, or other parameter. Once the controller **108** allows the suction valve **118** to move to the closed position while the pump **104** is operating, the suction valve **118** may resume “normal operation.” For example, under “normal operation” when the pump **104** moves the plunger **110** in the first direction **114**, the pressure in the pump chamber **140** may become lower than the pressure of the fluid in the intake manifold **122**, which causes the suction valve **118** to open allowing fluid through the suction valve **118** into the fluid end **112** from a fluid source **120**. When the plunger **110** begins to move in the second direction **116**, the suction valve **118** may close, preventing fluid from returning through the suction bore **132**. As such, the actuator **144** may be configured to retract the rod **146** such that the suction valve **118** is able to resume “normal operation” unimpeded by the unloading mechanism **142**.

(25) Additionally, or alternatively, the controller **108** may open the suction valve **118** once the driver speed of the prime mover **102** is below the threshold driver speed and the amount of time is a time between the driver speed passing below the threshold driver speed and the driver speed reaching a driver speed of approximately 0 revolutions per minute (RPM) (or other secondary threshold speed). Starting or winding down the prime mover **102** under a reduced load may significantly reduce fuel consumption, reduce electrical power requirements, allow the prime mover **102** to reach a rated speed and torque before being loaded, and may extend a life of the prime mover **102** and/or pump **104**, among other potential benefits.

(26) FIG. 2 depicts the example fluid system **100**. The fluid system **100** may be substantially the same as the fluid system **100** shown and described with respect to FIG. 1. However, as shown in FIG. 2, the controller **108** in FIG. 2 may be a manually operated and/or actuated controller and may be operable irrespective of the prime mover **102**. As such, the controller **108** may not need to be communicatively coupled to the prime mover **102** and/or sensors associated with the prime mover **102**. For example, the controller **108** may include a push button actuator, switch (e.g., pneumatic

switch, electronic switch, mechanical switch, etc.), lever, or other manually operated controller that is configured to control at least a portion of the operations of the fluid system **100**.

(27) In some examples, a user (such as an operator) may operate the controller **108** which may, in turn, be configured to operate the unloading mechanism **142** of the fluid system **100**. For example, a user may determine to operate the controller **108** while the prime mover **102** starts up, winds down, or during any other portion of operation of the fluid system **100**. In some examples, when the user operates (e.g., actuates a switch, presses a button, or otherwise operates and/or actuates the controller **108**) the controller **108** may cause one or more signals to be sent to the actuator **144** that is coupled to the rod **146** and the one or more signals may cause the actuator **144** to actuate the rod **146** between various positions. Additionally, or alternatively, when the user operates the controller **108**, the controller may be configured to manually engage the actuator **144** or the rod **146**, thereby causing the actuator **144** to actuate the rod **146** between various positions or to directly engage and move the rod **146** between various positions. Furthermore, in some examples, the controller **108** may be omitted and a user may directly control actuation **144** of the actuator via various input methods.

(28) FIG. 3 depicts a cross-sectional view of the fluid end **112** with the unloading mechanism **142** positioning the suction valve **118** in a closed position. As described previously, the fluid end **112** includes the block **130** having one or more bores formed in the block **130** of the fluid end **112**. For example, the fluid end **112** includes the suction bore **132** formed in the block **130** and defined along a first axis **300**. In some examples, the suction bore **132** provides a fluid passageway extending from an exterior surface of the block **130** to the pump chamber **140**. The suction bore **132** allows fluid to enter the fluid end **112** when the plunger **110** moves in the first direction **114**, allowing fluid into the pump chamber **140** of the fluid end **112**.

(29) The fluid end **112** includes a suction valve **118** disposed within the suction bore **132** that is configured to control fluid flow through the suction bore **132**. The suction bore **132** also includes a suction valve seat **302** disposed within the suction bore **132**. The suction valve seat **302** provides a surface against which the suction valve **118** rests when the suction valve **118** is positioned in a closed position **304**. When the suction valve **118** is in the closed position **304**, the suction valve **118** creates a fluid seal between the suction valve **118** and the suction valve seat **302**. As such, the suction valve **118** prevents fluid to flow from the pump chamber **140** of the fluid end **112** to the fluid intake manifold **122** when the suction valve **118** is positioned in the closed position **304**.

(30) The fluid end **112** includes a suction cover bore **306** formed in the block **130** of the fluid end **112** and extending along a second axis **308**. In some examples, the fluid end **112** includes a suction cover **312** and a suction cover retainer **310** that are disposed at least partially within the suction cover bore **306**. The fluid end **112** may include a spring **314** (or other biasing member) disposed between the suction valve **118** and a suction valve retainer **316**. The suction valve retainer **316** is configured to maintain a position of the spring **314** such that the spring exerts a force on the suction valve **118** to maintain the suction valve in the closed position **304**. In some examples, when the plunger **110** moves in the first direction **114**, the plunger **110** creates negative pressure within the pump chamber **140** that overcomes the biasing force exerted on the suction valve **118** by the spring **314**, thereby opening the suction valve **118**.

(31) As described previously, the system **100** includes an unloading mechanism **142** that is configured to allow the position of the suction valve **118** to be in the closed position **304** or an open position (as shown in FIG. 4). The unloading mechanism **142** may be configured to maintain the suction valve **118** in an open position while the prime mover **102** starts up, winds down, or otherwise operates. However, once the prime mover **102** reaches a threshold speed, the unloading mechanism **142** may allow the suction valve **118** to resume “normal operation,” allowing a pressure differential induced by the plunger **110**, as the plunger **110** moves in reciprocating motion, to open and close the suction valve **118**. The unloading mechanism **142** may include an actuator **144** coupled to a rod **146** and configured to move the rod **146** between various positions. As shown in

FIG. 3, the rod **146** is positioned in a retracted position (or a first position). In the retracted position, the rod **146** may be disposed proximate the suction valve **118** (e.g., being spaced therefrom) and/or may contact the suction valve **118**. In the retracted position, the suction valve **118** is positioned in the closed position **304**.

(32) In some examples, the actuator **144** may be coupled to an exterior surface of the fluid intake manifold **122** and the rod **146** may be coupled to the actuator **144** and may extend through an aperture **315** formed in the fluid intake manifold **122**. In some examples, the rod **146** may extend from the actuator **144** to a location proximate the suction valve **118**. In some examples, the rod **146** is coupled to the suction valve **118**. In some instances, the rod **146** includes an elongated rigid and/or semi-rigid rod. The rod **146** may comprise a metallic material such as aluminum, steel, stainless steel, or other metallic material.

(33) In some examples, the unloading mechanism **142** is communicatively coupled to the controller **108**. For example, the controller **108** may be communicatively coupled to the actuator **144** and is configured to control actuation of the actuator **144**. As such, the controller **108** may be configured to position the suction valve **118** in an open position or the closed position **304** via actuation of the actuator **144**, which moves the rod **146** between retracted position and an extended position (or a second position). In some examples, the controller **108** sends one or more signals to the actuator **144** to cause the actuator **144** to extend or retract the rod **146**, thereby causing the suction valve **118** to open or close.

(34) As described previously, the controller **108** receives driver speed data from the one or more sensors **106** of the prime mover **102**. The driver speed data may be indicative of a driver speed of the prime mover **102**. The controller **108** may be configured to control operation of the unloading mechanism **142** based at least in part on the driver speed data. For example, the controller **108** is configured to position the suction valve **118** in the closed position **304** when the driver speed data indicates that the driver speed is substantially equal to or greater than the threshold driver speed. In some instances, once the driver speed reaches the threshold driver speed, the prime mover **102** may be capable of driving operation of the pump **104** at a required torque, pump speed, pump rate, or other parameter. Once the controller **108** positions the suction valve **118** in the closed position while the pump **104** is operating, the suction valve **118** may resume “normal operation.” However, in some examples, the controller **108** may be manually operated by a user that operates the controller **108** to control operation of the unloading mechanism **142**. Furthermore, the controller **108** may receive additional sensor data from one or more other sensors of the system **100**. For example, the system **100** may receive pressure data associated with one or more components of the system, load data associated with the pump **104** and/or the prime mover **102**, and/or other data associated with the system **100**. The controller **108** may control operation of the unloading mechanism **142** based at least in part on the additional sensor data in addition to or instead of the driver speed data.

(35) The fluid end **112** further includes a discharge bore **134** formed in the block **130** and extending approximately along the first axis **300**. The discharge bore **134** forms a fluid passageway extending between an exterior surface of the block **130** and the pump chamber **140**. The discharge bore **134** allows fluid to be discharged from the fluid end **112** to a fluid discharge manifold **126** via an outlet **318** of the fluid end **112**. The fluid end **112** includes a discharge valve **124** disposed within the discharge bore **124** that is configured to control fluid flow through the discharge bore **134**. The discharge bore **134** also includes a discharge valve seat **320** disposed within the discharge bore **134**. In some examples, the discharge valve seat **320** provides a surface against which the discharge valve **124** rests when the discharge valve **124** is closed, creating a fluid seal between the discharge valve **124** and the discharge valve seat **320**. In some examples, when the plunger **110** moves in the second direction **116**, the plunger **110** displaces the fluid within the pump chamber **140** until a pressure of the fluid within the pump chamber **140** reaches and/or exceeds a threshold pressure, thereby forcing the discharge valve **124** to unseat (e.g., lift), opening the discharge valve **124**. The

plunger **110** directs the fluid out of the outlet **318** of the fluid end **112** and into a fluid discharge manifold **126** or other component.

(36) In some examples, the discharge bore **134** further includes a discharge cover **324** and a discharge cover retainer **322** that are disposed at least partially within the discharge bore **134**. The discharge cover **324** and the discharge cover retainer **322** cause fluid that flows through the discharge valve **124** to flow out of the fluid end **112** via the outlet **318**. The fluid end **112** may include a spring **326** (or other type of biasing member) that is disposed between the discharge valve **124** and the discharge cover **324**. The spring **326** is configured to maintain the discharge valve in a closed position **304** (as illustrated in FIG. 3) until the fluid within the pump chamber **140** reaches and/or exceeds a threshold pressure. That is, the spring **326** exerts a force on the discharge valve **124** in the closed position. Once the fluid is displaced to the threshold pressure, the fluid overcomes the biasing force exerted on the discharge valve **124** by the combined forces of the spring **326** and the existing pressure within the discharge bore **134**.

(37) The fluid end further includes the plunger bore **138** formed in the block **130** and extending approximately along the second axis **308** that may be substantially perpendicular to the first axis **300**. The plunger bore **138** is sized to receive the plunger **110** of the pump **104** at least partially within the plunger bore **110**. Furthermore, the plunger bore **138** is sized to allow the plunger **110** to move in the first direction **114** and the second direction **116** at least partially within the plunger bore **138** to allow fluid into the fluid end **112** via the suction bore **132**, displace the fluid in the pump chamber **140**, and discharge the fluid from the fluid end **112** via the discharge bore **134**.

(38) In some examples, the fluid intake manifold **122** may include a port **328** formed in the fluid intake manifold **122**. The port **328** may include a cap **330** that is configured to fluidly seal the port **328** when the cap **330** is coupled to the port **328**. In some examples, the cap **330** may be removed from the port **328** to allow fluid to drain out of the fluid intake manifold **122**.

(39) As mentioned previously, the unloading mechanism **142** may be configured to maintain the suction valve **118** in an open position while the prime mover **102** starts up, winds down, or otherwise operates. However, once the prime mover **102** reaches a threshold speed, the unloading mechanism **142** may allow the suction valve **118** to resume “normal operation,” allowing the plunger **110** to open and close the suction valve **118** as the plunger **110** moves in reciprocating motion. According to the present disclosure, the pump **104** (e.g., including the unloading mechanism **142**) may be operated independently of system pressure (e.g., fluid pressure downstream of the pump **104**). Therefore, the prime mover **102**, associated with the pump **104**, may be accelerated to an operating speed and operating torque against a reduced load that is less than an operating load that corresponds to system pressure. Starting or winding down the prime mover **102** under the reduced load may significantly reduce fuel consumption, reduce electrical power requirements, and may extend a life of the prime mover **102** and/or pump **104**, among other potential benefits.

(40) FIG. 4 depicts a cross-sectional view of the fluid end **112** with the unloading mechanism **142** positioning the suction valve **118** in an open position **402**. In some examples, the unloading mechanism **142** is configured to maintain the suction valve **118** in the open position **402** while the prime mover **102** starts up, winds down, or is otherwise operating below a threshold driver speed. Maintaining the suction valve **118** in the open position **402**, during operation of the pump **104**, opens two-way fluid circulation (e.g., through the suction bore **132**) between the pump chamber **140** and the fluid intake manifold **122** (and/or the fluid source **120**). For example, in contrast to normal operation, when the off-loading mechanism **142** causes the suction valve **118** to be positioned in the open position **402**, fluid may flow back from the pump chamber **140** to the fluid intake manifold **122** open position **402**. The unloading mechanism **142** may include an actuator **144** coupled to a rod **146** and configured to move the rod **146** between various positions. For example, the actuator **144** may extend the rod **146** which may press against the suction valve **118**, thereby lifting the suction valve **118** and positioning the suction valve **118** in the open position **402**.

(41) In some examples, the controller **108** receives driver speed data from the one or more sensors **106** of the prime mover **102**. The driver speed data is indicative of a driver speed of the prime mover **102**. The controller **108** may be configured to control operation of the unloading mechanism **142** based at least in part on the driver speed data. For example, the controller **108** may be configured to position the suction valve **118** in the open position **402** when the driver speed data indicates that the driver speed is below a threshold driver speed. However, in some examples, the controller **108** may be manually operated by a user that operates the controller **108** to control operation of the unloading mechanism **142**. In some instances, when the driver speed is below a threshold driver speed, the prime mover **102** may be starting up or winding down. As such, the controller **108** may open the suction valve **118** in order to unload the pump **104** for an amount of time necessary for the prime mover **102** to reach the threshold driver speed. For example, positioning the suction valve **118** in the open position **402** while the prime mover **102** starts up or winds down allows the prime mover **102** to start or wind down under a reduced load. In some examples, rather than having to pump fluid to a predetermined pressure to discharge fluid through the discharge valve **124**, the fluid is circulated between the pump chamber **140** and the fluid intake manifold **122** (and/or the fluid source **120**) as the plunger **110** moves in reciprocal motion within the plunger bore **138**. Starting or winding down the prime mover **102** under a reduced load may significantly reduce fuel consumption, reduce electrical power requirements, and may extend a life of the prime mover **102** and/or pump **104**, among other potential benefits. The controller **108** may be configured to allow the position of the suction valve **118** in the closed position **304** once the driver speed reaches and/or exceeds the threshold driver speed.

(42) FIG. 5 is a cross-sectional view of the fluid end **112** with an alternate example of the unloading mechanism **142** that is configured to position the suction valve **118**. The unloading mechanism **142** may be configured to maintain the suction valve **118** in an open position while the prime mover **102** starts up, winds down, or otherwise operates. However, once the prime mover **102** reaches a threshold speed, the unloading mechanism **142** may allow the suction valve **118** to resume “normal operation,” allowing the plunger **110** to open and close the suction valve **118** as the plunger **110** moves in reciprocating motion.

(43) In some examples, the unloading mechanism **142** shown in FIG. 5 may include an actuator **144**. In some examples, the actuator **144** may be coupled to an exterior surface of the fluid intake manifold **122**. Furthermore, the actuator **144** may be coupled to a surface of the fluid intake manifold **122** on a side of the fluid intake manifold **122** that is opposite the port **328** of the fluid intake manifold **122**. The actuator **144** may include an electronic actuator, a hydraulic actuator, or other type of actuator configured to actuate the rod **146**. The actuator **144** may be coupled to the rod **146** and is configured to move the rod **146** between various positions. As shown in FIG. 5, the rod **146** is positioned in an extended position. In the extended position, the rod **146** may be disposed proximate the suction valve **118** and/or may contact the suction valve **118**. In either instance, in the extended position, the rod **146** the suction valve **118** is positioned in the closed position **304**.

(44) In some instances, the rod **146** includes an elongated rigid and/or semi-rigid rod. The rod **146** may comprise a metallic material such as aluminum, steel, stainless steel, or other metallic material. As shown in FIG. 5, the rod **146** may include one or more bends **500** formed in the rod **146** and/or the rod **146** may be formed from multiple sections that are coupled together. In either example, the rod **146** extends from the actuator **144** through an aperture **502** formed in the fluid intake manifold **122** and to a location proximate and/or in contact with the suction valve **118**.

(45) In some examples, a portion of the rod **146** is secured within a joint **504**. The joint **504** may be disposed at least partially within the aperture **502** in the fluid intake manifold **122**. The joint **504** may allow the rod **146** to move in one or more directions within the joint **504**. For example, the joint **504** may allow the rod **146** to move rotationally around a central axis **506** of the joint **504**. The joint **504** may include a pivot joint, ball joint, or other type of joint. In some examples, the joint **504**

provides a pivot point about which the rod **146** is moveable when the actuator **144** causes movement of the rod **146**.

(46) In some examples, the unloading mechanism **142** is communicatively coupled to the controller **108**. For example, the controller **108** may be communicatively coupled to the actuator **144** and is configured to control actuation of the actuator **144**. As such, the controller **108** may be configured to position the suction valve **118** in an open position **402** (as illustrated in FIG. 6) or the closed position **304** via actuation of the actuator **144**, which moves the rod **146** between retracted position and an extended position (or a second position). In some examples, the controller **108** sends one or more signals to the actuator **144** which causes the actuator **144** to extend or retract the rod **146**, thereby causing the suction valve **118** to open or close. However, in some examples, the controller **108** may be manually operated by a user that operates the controller **108** to control operation of the unloading mechanism **142**.

(47) As described previously, the controller **108** receives driver speed data from the one or more sensors **106** of the prime mover **102**. The driver speed data is indicative of a driver speed of the prime mover **102**. The controller **108** may be configured to control operation of the unloading mechanism **142** based at least in part on the driver speed data. For example, the controller **108** is configured to allow the position of the suction valve **118** in the closed position **304** when the driver speed data indicates that the driver speed is substantially equal to or greater than the threshold driver speed. However, in some examples, the controller **108** may be manually operated by a user that operates the controller **108** to control operation of the unloading mechanism **142**. In some instances, once the driver speed reaches the threshold driver speed, the prime mover **102** may be capable of driving operation of the pump **104** at a required torque, pump speed, pump rate, or other parameter. Once the controller **108** positions the suction valve **118** in the closed position while the pump **104** is operating, the suction valve **118** may resume “normal operation.” Starting or winding down the prime mover **102** under a reduced load may significantly reduce fuel consumption, reduce electrical power requirements, and may extend a life of the prime mover **102** and/or pump **104**, among other potential benefits.

(48) FIG. 6 depicts a cross-sectional view of the fluid end **112** with the unloading mechanism **142** positioning the suction valve in the open position **402**. In some examples, the unloading mechanism **142** is configured to maintain the suction valve **118** in an open position while the prime mover **102** starts up, winds down, or is otherwise operating below a threshold driver speed. Maintaining the suction valve **118** in the open position while the pump **104** operates permits fluid to circulate between the pump chamber **142** and the fluid intake manifold **122** (and/or the fluid source **120**). As shown in FIG. 6, the unloading mechanism **142** includes the actuator **144** coupled to the rod **146** and configured to move the rod **146** between various positions. For example, the actuator **144** may retract the rod **146** which may cause the rod **146** to press against the suction valve **118**, thereby lifting the suction valve **118** and positioning the suction valve **118** in the open position **402**. As mentioned previously, when the actuator **144** moves the rod **146**, the rod **146** may rotate around the joint **504**, thereby causing the rod **146** to press against the suction valve **118**.

(49) In some examples, the controller **108** receives driver speed data from the one or more sensors **106** of the prime mover **102**. The driver speed data is indicative of a driver speed of the prime mover **102**. The controller **108** may be configured to control operation of the unloading mechanism **142** based at least in part on the driver speed data. For example, the controller **108** may be configured to position the suction valve **118** in the open position **402** when the driver speed data indicates that the driver speed is below a threshold driver speed. However, in some examples, the controller **108** may be manually operated by a user that operates the controller **108** to control operation of the unloading mechanism **142**. In some instances, when the driver speed is below a threshold driver speed, the prime mover **102** may be starting up or winding down. As such, the controller **108** may open the suction valve **118** in order to unload the pump **104**. For example, positioning the suction valve **118** in the open position **402** while the prime mover **102** starts up or

winds down allows the prime mover **102** to start or wind down under a reduced load. For example, rather than having to pump fluid to a predetermined pressure to discharge fluid through the discharge valve **124**, the fluid is circulated between the pump chamber **140** and the fluid intake manifold **122** (and/or the fluid source **120**) as the plunger **110** moves in reciprocal motion within the plunger bore **138**. Starting or winding down the prime mover **102** under a reduced load may significantly reduce fuel consumption, reduce electrical power requirements, and may extend a life of the prime mover **102** and/or pump **104**, among other potential benefits.

(50) FIG. 7 illustrates an exemplary method for unloading a prime mover **102** configured to drive operation of a pump **104**. The example method is illustrated as a collection of steps in a logical flow diagram, which represents operations that may be implemented in hardware, software, or a combination thereof. In the context of software, the steps represent computer-executable instructions stored in memory. Such computer-executable instructions may include routines, programs, objects, components, data structures, and the like that perform particular functions or implement particular abstract data types. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described steps may be combined in any order and/or in parallel to implement the process. For discussion purposes, and unless otherwise specified, the method **700** is described with reference to the system **100**, the prime mover **102**, the pump **104**, the one or more sensors **106**, the controller **108**, the fluid end **112** and the components thereof, and the unloading mechanism **142**. In particular, and unless otherwise specified, the method **700** will be described below as being performed by the controller **108** for ease of description. Additionally, and/or alternatively, at least a portion of the method **700** may be performed by user input via the controller **108** and/or one or more user interfaces included on various components of the system **100**. As mentioned previously, a user (such as an operator) may operate the controller **108** which may, in turn, be configured to operate an unloading mechanism **142** of the fluid system **100**.

(51) With reference to FIG. 7, at **702**, the method **700** includes determining to start the prime mover **102**. In some examples, a user may determine to start the prime mover **102** and may cause the prime mover **102** to start up. Additionally, or alternatively, the controller **108** (or another controller of the fluid system **100** such as a dedicated prime mover ECM, ECU, or other controller) receives instructions to start the prime mover **102**. In some examples, a user may provide user input via a switch, user interface, or other selectable control that provides instructions to the controller **108** to start the prime mover **102**. Additionally, or alternatively, the controller **108** may receive instructions to start the prime mover from a computing device that is configured to control at least a portion of a hydraulic fracturing process.

(52) At **704**, the method **700** includes causing the suction valve **118** to be positioned in the open position **402**. In some examples, a user may operate or otherwise provide input to the controller **108** which causes the suction valve **118** to be positioned in the open position **402**. In some examples, the controller **108** causes the suction valve **118** to be positioned in the open position **402** based at least in part on receiving instructions to start the prime mover **102**. In some instances, causing the suction valve **118** to be positioned in the open position **402** includes sending, via the controller **108**, one or more signals to the actuator **144**. However, in some examples, the user may manually operate (e.g., via a switch, button, lever, or other control) the controller **108** to open the suction valve **118**. As described previously, the actuator **144** is configured to actuate the rod **146** between various positions and the one or more signals may cause the actuator **144** to position the rod in a first position or a second position. In the first position, the rod **146** is disposed proximate and/or in contact with the suction valve **118** and the suction valve **118** is in the closed position **304**. In the second position, the actuator **144** actuates the rod **146** to press against the suction valve **118** positioning the suction valve **118** in the open position **402**. As such, when the controller **108** positions the suction valve **118** in the open position **402**, the controller **108** may send one or more signals to the actuator **144** causing the actuator **144** to actuate the rod **146** into the second position,

which causes the rod **146** to abut the suction valve **118**, thereby causing the suction valve **118** to be positioned in the open position **402**.

(53) At **706**, the controller **108** receives one or more signal from the fluid system **100**. For example, the controller **108** may receive one or more signals indicative of driver speed data from the one or more sensors **106** of the prime mover **102**. In some examples, the driver speed data represents a rotational speed of a drive shaft of the prime mover **102** and may be represented in revolutions per minute (RPM). However, in some examples, the one or more signals represent one or more parameters associated with vibration (associated with the fluid end **112**, the pump **104**, and/or the prime mover **102**), pressure (associated with the fluid end **112** and/or the pump **104**), or one or more other parameters associated with the fluid system **100**.

(54) At **708**, the controller **108** determines whether operation of the fluid system **100** satisfies a threshold. For example, the controller **108** may determine whether the driver speed of the prime mover **102** is substantially equal to or greater than a threshold driver speed. As described herein, “substantially equal to” may mean that the driver speed is within: approximately 50 RPM of the threshold driver speed, approximately 100 RPM of the threshold driver speed, approximately 250 RPM of the threshold driver speed, or approximately 500 RPM of the threshold driver speed. Additionally, and/or alternatively, the controller **108** may determine, at **708**, whether one or more of the parameters previously mentioned (e.g., vibration, pressure, or other parameters) satisfy a threshold value. In some examples, the controller **108** may determine whether multiple parameters satisfy multiple respective thresholds and may control operation (either automatically, semi-automatically, or manually via user input) of the unloading mechanism based on determining whether the multiple parameters satisfy their respective threshold values.

(55) If the controller **108** determines at **708** that operation of the fluid system **100** does not satisfy the threshold (Step: **708**—No), the process **700** returns to **704** and the controller **108** maintains the open position **402** of the suction valve **118**. In some examples, the controller **108** may continue to receive one or more signals from the fluid system **100** at **706** and determine whether the operation of the fluid system **100** satisfies the threshold at **708**.

(56) If, however, the controller **108** determines at **708** that operation of the fluid system **100** satisfies the threshold (Step: **708**—Yes), the process **700** proceeds to **710** where the method **700** includes causing the suction valve **118** to be positioned in the closed position **304**. In some examples, causing the suction valve **118** to be positioned in the closed position **304** includes sending, via the controller **108**, one or more signals to the actuator **144**. In some examples, a user may operate or otherwise provide input to the controller **108** which causes the suction valve **118** to be positioned in the closed position **304**. In either example, the controller **108** may cause the actuator **144** to actuate the rod **146** to position the rod **146** in the first position. In the first position, the rod **146** is disposed proximate and/or in contact with the suction valve **118** and the suction valve **118** is positioned in the closed position **304**. As described previously, when the rod **146** is positioned in the first position, the suction valve **118** resumes “normal operation” as the pump **104** continues to operate.

(57) As shown in FIG. 7, the controller **708** may continue to receive one or more signals from the fluid system **100** once the suction valve **118** is positioned in the closed position **304**. The controller **108** may continue to monitor operation of the fluid system **100** and determine whether the operation of the fluid system **100** satisfies a threshold at **708**. The controller **108** may also position the suction valve **118** in the open position if the fluid system **100** fails to satisfy the threshold.

INDUSTRIAL APPLICABILITY

(58) The present disclosure provides a system and mechanism for unloading a pump during start up or wind down of a prime mover that drives the pump. The system can be used in a variety of applications. For example, the system is used in gas, oil, and hydraulic fracturing applications. The system includes an unloading mechanism that maintains an open position of a valve within a fluid end to unload a pump during start up or wind down of the prime mover that drives the pump.

Starting or winding down the prime mover with the valve open in the fluid end reduces a load imparted on the prime mover as the prime mover drives the pump. Furthermore, starting or winding down the prime mover under a reduced load can significantly reduce fuel consumption, reduce electrical power requirement and/or can extend a life of the prime mover and/or the pump.

(59) According to some embodiments, the system **100** includes a controller **108** configured to position a suction valve **118** in an open position **402** or a closed position **304**. The controller **108** positions the suction valve **118** in the open position **402** or the closed position **304** while a prime mover **102** and a pump **104** are operating. The controller **108** is communicatively coupled to an unloading mechanism **142** that positions the suction valve **118** in a position. The unloading mechanism **142** includes an actuator **144** that actuates a rod **146** between various positions in order to lift the suction valve **118** or allow the suction valve **118** to be seated on a suction valve seat **302**. The controller **108** is configured to position the suction valve **118** in the open position **402** when the driver speed is below a threshold driver speed and to position the suction valve in the closed position **304** when the driver speed is substantially equal to or greater than the threshold driver speed.

(60) While aspects of the present disclosure have been particularly shown and described with reference to the embodiments above, it will be understood by those skilled in the art that various additional embodiments may be contemplated by the modification of the disclosed machines, systems and methods without departing from the spirit and scope of what is disclosed. Such embodiments should be understood to fall within the scope of the present disclosure as determined based upon the claims and any equivalents thereof.

Claims

1. A system, comprising: a fluid end including: a block; a suction bore formed in the block; a suction valve seat disposed within the suction bore; a suction valve disposed within the suction bore, the suction valve, in a closed position, being configured to contact the suction valve seat to seal the suction bore; and a rod that is moveable between a first position and a second position; and a controller configured to: determine, based at least in part on driver speed data indicative of a driver speed of a prime mover coupled to a hydraulic fracturing pump that includes the system, whether the driver speed, of the prime mover coupled to the hydraulic fracturing pump, satisfies a threshold; cause, when the driver speed, of the prime mover coupled to the hydraulic fracturing pump, does not satisfy the threshold, hydraulic fracturing fluid to flow from a pump chamber of the hydraulic fracturing pump to a fluid intake manifold of the hydraulic fracturing pump by causing the rod to engage the suction valve in the second position; and cause the rod to retract, into the first position, when the driver speed satisfies the threshold, wherein: the rod is configured to be in the first position when the suction valve is in the closed position, and the rod is configured to be in the second position when engagement of the rod with the suction valve causes the suction valve to be in an open position.

2. The system of claim 1, further comprising an actuator configured to actuate the rod between the first position and the second position.

3. The system of claim 2, wherein the controller is communicatively coupled to the actuator, and wherein the controller is configured to cause the actuator to position the rod in the second position.

4. The system of claim 1, wherein the fluid end further includes a plunger bore formed in the block, wherein the system comprises: the hydraulic fracturing pump; and the prime mover, the hydraulic fracturing pump having a plunger disposed at least partially within the plunger bore, the prime mover being configured to drive operation of the hydraulic fracturing pump, and the prime mover including a driver speed sensor configured to generate the driver speed data, and wherein the controller is communicatively coupled to the prime mover and the driver speed sensor, and wherein the controller is further configured to receive the driver speed data.

5. The system of claim 1, wherein, to cause the rod to retract, the controller is configured to: determine that the driver speed, of the prime mover coupled to the hydraulic fracturing pump, is equal to or greater than the threshold; and cause the rod to be positioned in the first position based at least in part on determining that the driver speed, of the prime mover coupled to the hydraulic fracturing pump, is equal to or greater than the threshold.
6. The system of claim 1, wherein the fluid end further comprises: one or more other suction valves; and one or more other rods that correspond to the one or more other suction valves.
7. The system of claim 6, further comprising: multiple actuators corresponding to the rod and the one or more other rods.
8. The system of claim 7, wherein the controller is configured to control the multiple actuators.
9. The system of claim 1, wherein the fluid end further includes one or more other suction valves, and wherein the controller is configured to close or open the suction valve and the one or more other suction valves in a specific sequence in order to reduce a rate of load being applied to the prime mover.
10. A method, comprising: determining, based at least in part on driver speed data indicative of a driver speed of a prime mover coupled to a fracturing pump, whether the driver speed, of the prime mover coupled to the fracturing pump, satisfies a threshold; causing, when the driver speed, of the prime mover coupled to the fracturing pump, does not satisfy the threshold, fracturing fluid to circulate between one or more respective pump chambers of the fracturing pump and a fluid manifold by positioning one or more suction valves in an open position; and causing, when the driver speed satisfies the threshold, the one or more suction valves to be positioned in a closed position, wherein the one or more suction valves prevent flow from the one or more respective pump chambers to the fluid manifold when the one or more suction valves are in the closed position.
11. The method of claim 10, wherein positioning the one or more suction valves in the open position includes: causing, via a controller, one or more actuators to position one or more rods to engage the one or more suction valves to position the one or more suction valves in the open position.
12. A system comprising: a fracturing pump; a prime mover coupled to the fracturing pump; and a controller configured to: determine, based at least in part on driver speed data indicative of a driver speed of the prime mover, whether the driver speed satisfies a threshold; and cause fracturing fluid to flow from a pump chamber of the fracturing pump to a fluid intake manifold of the fracturing pump by causing a suction valve, of the fracturing pump, to be positioned in an open position for an amount of time until the driver speed satisfies the threshold, the suction valve being configured to be substantially static in the open position for the amount of time; and allow the suction valve to be positioned in a closed position after the driver speed satisfies the threshold.
13. The system of claim 12, further comprising: a rod disposed adjacent the suction valve; and an actuator configured to actuate the rod, wherein the rod is configured to press against the suction valve to position the suction valve in the open position.
14. The system of claim 12, wherein the amount of time is a time between starting the prime mover and the driver speed reaching the threshold.
15. The system of claim 12, further comprising: a rod disposed adjacent the suction valve, wherein, allowing the suction valve to be positioned in the closed position, the controller is configured to: cause the rod to position the suction valve in the closed position when the driver speed is equal to or greater than the threshold.
16. The system of claim 12, wherein the prime mover is a diesel engine or a natural gas engine.
17. The system of claim 12, wherein the prime mover is configured to drive operation of the fracturing pump at a required torque or a required speed when the driver speed satisfies the threshold.
18. The system of claim 1, wherein the driver speed, of the prime mover coupled to the hydraulic

fracturing pump, is a rotational speed of a drive shaft of the prime mover coupled to the hydraulic fracturing pump, and wherein, to determine whether the driver speed, of the prime mover coupled to the hydraulic fracturing pump, satisfies the threshold, the controller is further configured to: determine whether the driver speed, of the prime mover coupled to the hydraulic fracturing pump, is within 50 revolutions per minute (RPM) of the threshold.

19. The system of claim 1, wherein, to cause the rod to retract, into the first position, when the driver speed satisfies the threshold, the controller is further configured to: cause the rod to retract, into the first position, based on determining that the driver speed satisfies the threshold, wherein the controller is further configured to: determine that the driver speed, of the prime mover coupled to the hydraulic fracturing pump, does not satisfy the threshold after determining that the driver speed, of the prime mover coupled to the hydraulic fracturing pump, satisfies the threshold and after causing the rod to retract into the first position, and wherein, to cause, when the driver speed, of the prime mover coupled to the hydraulic fracturing pump, does not satisfy the threshold, the hydraulic fracturing fluid to flow from the pump chamber of the hydraulic fracturing pump to the fluid intake manifold of the hydraulic fracturing pump, the controller is further configured to: cause, after causing the rod to retract into the first position and based on determining that the driver speed, of the prime mover coupled to the hydraulic fracturing pump, does not satisfy the threshold, the hydraulic fracturing fluid to flow from the pump chamber of the hydraulic fracturing pump to the fluid intake manifold of the hydraulic fracturing pump.

20. The method of claim 10, further comprising: causing the one or more suction valves to be positioned in the open position after causing the one or more suction valves to be positioned in the closed position.
